Chinese-SkillSpan: A Span-Level Dataset for ESCO-Derived Competency Extraction from Chinese Job Ads

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ABSTRACT

Job Skill Named Entity Recognition (JobSkillNER) identifies textual spans that express job-related competencies in recruitment text. Progress in Chinese JobSkillNER is limited by the absence of a large, documented span-level resource with Chinese-specific boundary and type rules. We introduce Chinese-SkillSpan, a corpus of 22,840 sentences sampled from four heterogeneous Chinese recruitment sources. Its flat annotation scheme distinguishes language competence, knowledge, specific or operational skills, and transversal competence (LSKT), following high-level competency distinctions used in ESCO. Chinese-specific operational guidelines define complete, boundary-consistent, and non-overlapping spans. The construction workflow combines language-model-generated annotation drafts with human validation, conflict adjudication, and quality control. The benchmark provides an identifier-aligned standardized reference, a reproducible scorer, and diagnostic analyses across source domains, span lengths, and boundary conventions. Baseline experiments show that exact-span performance is sensitive to Chinese boundary decisions and domain variation, which remain challenging for both pretrained encoders and large language models. Historical protocol results and internal artifact identifiers are confined to the supplementary reproducibility record. Project materials are maintained at `https://sites.google.com/view/cn-skillspan-resources`.

INTRODUCTION

Online recruitment platforms provide large volumes of job advertisements that can support timely analyses of labor-market demand and skill requirements (International Labour Organization, 2020). A central task in this setting is Job Skill Named Entity Recognition (JobSkillNER), which identifies textual spans expressing job-related competencies in unstructured job descriptions. Reliable JobSkillNER supports downstream applications such as skill profiling, job recommendation, curriculum planning, and aggregate labor-market intelligence.

We use **LSKT** to denote four complementary competency categories: **L**anguage competence, **K**nowledge, **S**pecific or operational skills, and **T**ransversal competence. English resources such as SkillSpan demonstrate the value of expert-reviewed span annotations for boundary-sensitive skill extraction (Zhang et al., 2022b). However, publicly available Chinese recruitment resources do not yet combine span-level annotation, an ESCO-derived LSKT type system, multiple recruitment domains, and an auditable evaluation protocol. This gap limits reproducible model development and cross-lingual comparison.

Constructing such a resource is challenging for three reasons. First, Chinese text does not mark word boundaries explicitly, so exact-span annotation is sensitive to segmentation and to whether modifiers, objects, or complements are retained. Second, compressed expressions and coordination structures create recurrent ambiguity between knowledge and executable skills, and

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between transversal competences and task-specific behavior. Third, recruitment sources differ substantially in genre and vocabulary, making a convention learned from one source difficult to transfer unchanged to another. Large-scale manual annotation is also costly; language-model-assisted drafting can improve efficiency, but its outputs require human verification because model suggestions may anchor later decisions (Yang et al., 2025; Oliveira et al., 2024).

To address these challenges, we introduce **Chinese-SkillSpan**, a Chinese span-level JobSkillNER resource containing 22,840 sentences from four heterogeneous recruitment sources. Its four LSKT categories are derived from high-level competency distinctions in ESCO (Commission, 2024b). Language models generate annotation drafts at scale, after which human annotators verify span boundaries and labels, resolve conflicts, and perform quality checks. The draft-generation prompt and human handbook share the same Chinese-specific rules for complete spans and contextual type decisions; machine suggestions never override human-selected source-text offsets. Agreement is reported using span-level F1 and Cohen's κ to make the remaining annotation uncertainty explicit (Landis and Koch, 1977).

The benchmark centers on one standardized annotation and evaluation protocol. Its current 980-record verification subset is undergoing final manual review, so all pre-release results will be recomputed after adjudication and freezing. Results under earlier span conventions are retained only in the appendix for provenance and protocol-sensitivity analysis, rather than presented as a competing leaderboard. Exact file versions, internal identifiers, coverage rules, and scorer metadata are recorded in the reproducibility inventory and public repository. Overall, this work makes the following contributions:

- We release **Chinese-SkillSpan**, a 22,840-sentence Chinese JobSkillNER resource covering four recruitment sources and four ESCO-derived LSKT categories.
- We provide Chinese-specific operational guidelines and an auditable annotation workflow that combines language-model-generated drafts with human boundary verification, label adjudication, and quality control.
- We define an identifier-aligned standardized benchmark with explicit prediction coverage, exact and relaxed span metrics, frozen manifests, and strict separation between the released reference and historical analyses.
- We characterize the benchmark through encoder and language-model baselines, source-domain analysis, span-length analysis, multi-seed stability, and cross-protocol boundary diagnostics.

Figure 1 summarizes the separation between benchmark construction, model training, and held-out evaluation.

DATASET AND ANNOTATION FRAMEWORK

Task and Label Space

Given a Chinese job-ad sentence, JobSkillNER returns a set of flat, non-overlapping spans. Each span is represented by its character offsets and one label from the LSKT inventory: **L** for language competence, **K** for knowledge, **S** for specific or operational skills, and **T** for transversal competence. The annotation guidelines use *minimal-complete* boundaries: the selected substring is as short as possible while remaining a self-contained competency mention, and modifiers are retained only when they change the competency expressed by the span.

The four-way inventory is *derived from* high-level distinctions used in ESCO-1.20. The present benchmark is limited to ESCO-derived LSKT span extraction: it evaluates boundary detection and LSKT type assignment.

Operational Span Annotation Protocol

The standardized annotation protocol preserves the four labels L, K, S, and T throughout annotation; labels are not collapsed when spans are created. Any two-class projection is computed only as a secondary evaluation view. Each span must be an exact contiguous substring of the

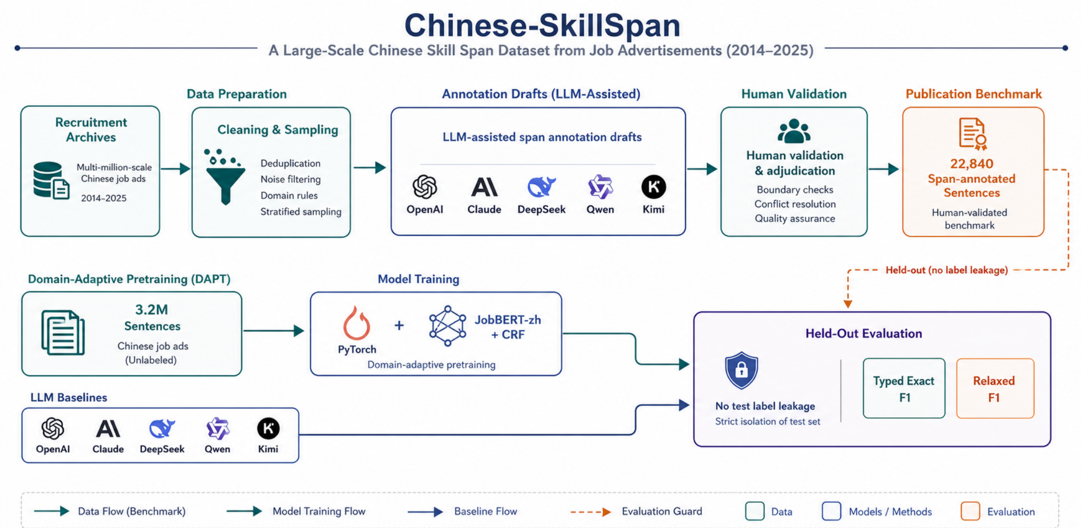

Figure 1. Overview of Chinese-SkillSpan data construction, language-model-assisted annotation, human validation, domain-adaptive pretraining, and held-out evaluation. The dashed boundary indicates that held-out test labels are isolated from model training, prompt and model selection, and hyperparameter tuning.

original sentence, cover the shortest complete mention rather than a truncated word fragment, and remain flat and non-overlapping. The preferred span length is two to eight tokens under the documented segmentation view, but this preference is not a hard cap and never licenses semantic truncation. An entire job-responsibility clause is not labeled as a single S span. Canonical names and ontology links, when available, are stored separately from the extracted surface form. Internal version identifiers are confined to the reproducibility inventory and do not denote different prediction tasks.

For constructions headed by expressions meaning *familiar with*, *master*, *proficient in*, *understand*, or *possess*, annotators retain the competency object but exclude the trigger. Sentences limited to application logistics, medical examinations, public notices, employee benefits, motivational boilerplate, or an explicitly negated requirement receive no positive competency span. Natural-language names, proficiency levels, and language examinations—for example, English, CET-6, or JLPT N2—are labeled L, whereas technical standards and professional certifications are labeled K. Programming languages, frameworks, software tools, and platforms required for job use are labeled S; explicit theory or principles are labeled K. Thus, in “使用 SQL 进行报表制作”, *SQL* and 报表制作 are separate S spans, whereas *SQL* 原理 is K. No global label-priority rule is used: unresolved K/S, L/S, L/K, or T/S cases are adjudicated with a recorded rationale.

Corpus and Annotation Layers

The processed corpus contains 22,840 sentences drawn from four recruitment sources and is partitioned into 17,460 training, 2,143 development, and 3,237 test sentences. At the scale-oriented drafting stage (Macro), language models generate candidate spans and labels. At the record-level verification stage (Micro), human annotators use Doccano, a web-based sequence-labeling interface, to inspect boundaries and types, resolve conflicts, and perform quality checks. We use *Silver* for machine-drafted annotations that have not completed the specified human review, and *Gold* only for a frozen human-reviewed reference. Record-level provenance is retained for any mixed-provenance diagnostic artifact.

The standardized benchmark reference contains 2,601 identifier-aligned test records inherited from the same source sentences used in an earlier human-review round. The span convention was subsequently standardized through the operational rules above, followed by renewed human verification and adjudication. Results under the earlier convention are retained only in the appendix for provenance and sensitivity analysis.

For the encoder experiments, we fine-tune the existing JobBERT-zh encoder (RoBERTa-wwm-ext with job-description masked-language-model adaptation) and a CRF head using protocol-aligned training annotations and the 1M and 3M domain-adapted checkpoints. Human-selected, zero-based, end-exclusive character offsets remain authoritative. Deterministic Chinese word segmentation is used to flag possible mid-word boundaries and to construct a separately named boundary-aligned evaluation view; it does not silently overwrite a frozen human reference. Every reported derived view freezes the segmenter version, user dictionary, transformation function, token cap, and checksum and applies the same named transformation to reference annotations and predictions.

The current pre-release reference uses all 2,601 identifiers. A 980-record subset is undergoing final manual verification under the standardized protocol, whereas the other 1,621 records retain protocol-aligned draft labels with explicit provenance. Consequently, the current reference is not described as final Gold, and all reported scores are provisional. After verification and adjudication, every benchmark result will be recomputed from a single frozen manifest. The human-frozen character-offset file will remain unchanged, while any segmentation-aligned file will be released as a named derivative. Results use identifier-strict alignment, typed exact micro-F1, and typed relaxed micro-F1 at IoU ≥ 0.5. Current language models are evaluated with the same extraction prompt, which is synchronized with the human handbook. Exact artifact names, internal protocol versions, prompt hashes, and legacy references are preserved in the appendix inventory and public repository.

Boundary and Type Decisions

The guidelines address three recurring sources of ambiguity in Chinese recruitment text. First, coordinated expressions are split only when each conjunct independently denotes a competency; conjuncts that share an indispensable head or complement remain one complete span. Second, contextual words such as degree adverbs and generic ability markers are excluded unless they alter the competency meaning. Third, spans that plausibly fit more than one LSKT category are resolved using a contextual decision table rather than model confidence or a global label-priority rule. Following the guideline-development principles used in SkillSpan (Zhang et al., 2022b), the release protocol separates calibration, independent overlap annotation, adjudication, and post-revision rechecking. The supplementary materials provide the operational rules, the rule-change history, type decisions, annotation schema, audit workflow, and worked examples.

Release and Audit Protocol

Each released benchmark version is required to include a data manifest, file checksums, a schema, the annotation guidelines, and a scorer. In strict evaluation mode, the scorer aligns records by sample identifier and rejects missing, extra, or duplicate identifiers. The primary metric is typed exact-span micro-F1; collapsed exact-span and relaxed-overlap scores are secondary diagnostics. This protocol is designed to make coverage errors visible rather than silently omitting unmatched records.

Appendix Table 12 maps the standardized benchmark and each historical evaluation layer to its frozen artifact, scorer, and permitted reporting role. This mapping prevents results produced under different annotation conventions from being merged into a single leaderboard.

Table 1 reports corpus and span statistics by split and source, while Figure 2 summarizes the corresponding span-length distributions. The complete processed test split contains 3,237 sentences; Test-V4 is the identifier-aligned 2,601-record pre-release reference described above.

EXPERIMENTS

Evaluation Setup

The processed test split contains 3,237 sentences. The standardized benchmark evaluates 2,601 identifier-aligned test records drawn from this split. Of these records, 980 are undergoing final human verification and adjudication, while the remaining 1,621 retain protocol-aligned draft annotations with explicit record-level provenance. The results reported in this section are therefore pre-release estimates. Every result will be recomputed from the frozen human-adjudicated

Table 1. Statistics of Chinese-SkillSpan. Training and development spans follow LSKT-v4 silver labels; Test-V4 is the mixed-provenance matched pre-release reference, with a 980-record SimHuman rule overlay and Jieba alignment. AI, Cloud, and Public are `source_domain` values; Grad denotes graduate recruitment and appears only in training and development. Overlap is zero because labels are flat.

	Statistics ↓ / src. →	AI	Cloud	Public	Grad	Total
Train (v4 silver)	# Posts	630	—	—	970	1,600
	# Sentences	7,148	—	—	10,312	17,460
	# Tokens	261,788	—	—	351,163	612,951
	# Skill spans (S+T)	9,780	—	—	8,864	18,644
	# Knowledge spans (L+K)	4,047	—	—	2,562	6,609
	# L	77	—	—	262	339
	# K	3,970	—	—	2,300	6,270
	# S	7,357	—	—	5,058	12,415
	# T	2,423	—	—	3,806	6,229
	# Sentences with overlap	0	—	—	0	0
Dev (v4 silver)	# Posts	200	—	—	—	200
	# Sentences	2,143	—	—	—	2,143
	# Tokens	79,783	—	—	—	79,783
	# Skill spans (S+T)	2,967	—	—	—	2,967
	# Knowledge spans (L+K)	1,411	—	—	—	1,411
	# L	32	—	—	—	32
	# K	1,379	—	—	—	1,379
	# S	2,244	—	—	—	2,244
	# T	723	—	—	—	723
	# Sentences with overlap	0	—	—	—	0
Test-V4 (pre-release)	# Posts	140	40	20	—	200
	# Sentences	1,407	457	737	—	2,601
	# Tokens	56,725	16,712	34,050	—	107,487
	# Skill spans (S+T)	2,051	563	174	—	2,788
	# Knowledge spans (L+K)	973	257	173	—	1,403
	# L	22	10	1	—	33
	# K	951	247	172	—	1,370
	# S	1,535	425	130	—	2,090
	# T	516	138	44	—	698
	# Sentences with overlap	0	0	0	—	0
DAPT	# Sentences	409,400	—	—	590,600	1,000,000
	# Sentences (3.2M mix)	1,310,080	—	—	1,889,920	3,200,000

reference before submission, and the current values must not be interpreted as final Gold performance.

Predictions are aligned by identifier; missing, extra, or duplicate identifiers are treated as coverage failures rather than silently omitted. The primary metric is typed exact-span micro-F1, for which a true positive requires identical character boundaries and the same LSKT label. Typed relaxed micro-F1 uses an intersection-over-union threshold of $\mathrm{IoU} \geq 0.5$ and is reported as a secondary measure of semantic and boundary overlap. Human-selected character offsets remain authoritative. The deterministic word-boundary alignment used in the current analysis is applied symmetrically to the reference view and predictions and is released as a separately named transformation.

The encoder baselines combine the Chinese JobBERT-zh representation with a CRF prediction head and protocol-aligned training annotations. The language-model baselines receive the same standardized extraction prompt without access to test labels. Exact model identifiers, access dates, prompt hashes, transformation settings, historical references, and file checksums are maintained in the reproducibility inventory and repository rather than used as narrative protocol names in the main text.

Annotation Reliability Status

The standardized reference is still undergoing human verification. Before it is frozen, two annotators will independently label a stratified 300-record subset from raw text without seeing model drafts, earlier spans, or each other's work. Typed exact and relaxed span scores, character-level Cohen's κ over O/L/K/S/T, and per-label results will be recorded before third-annotator adju-

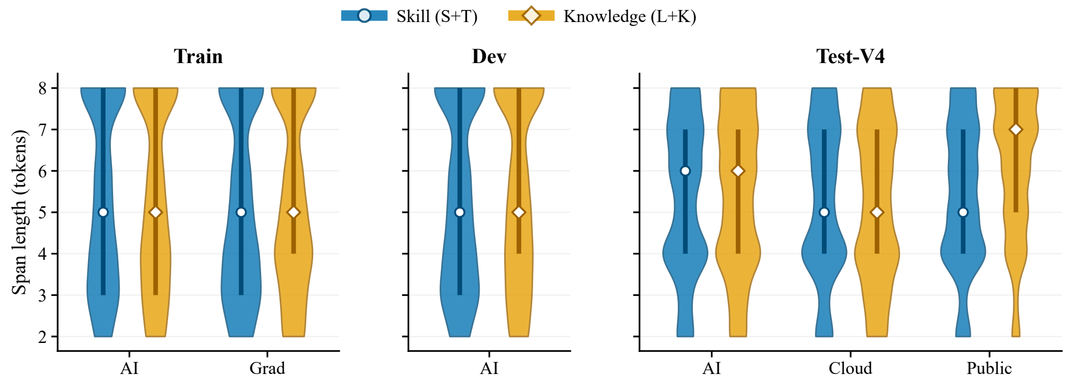

Figure 2. Span-length distributions following the layout of SkillSpan Figure 2. Train and development use LSKT-v4 silver labels. Test-V4 uses the mixed-provenance matched pre-release reference on the same 2,601 identifiers, including the 980-record SimHuman rule overlay; it is not a frozen human-adjudicated Gold release. Skill denotes S+T and Knowledge denotes L+K. Full violins show the complete distributions; white markers denote medians and thick vertical segments denote interquartile ranges. Development contains the AI source only, whereas graduate recruitment appears in training only.

dication. No agreement threshold or reliability value is claimed in advance. Agreement statistics from the earlier annotation convention are reported only as historical provenance in Appendix Table 14.

Representative Annotations

Figure 3 provides representative examples of the four labels and their span boundaries.

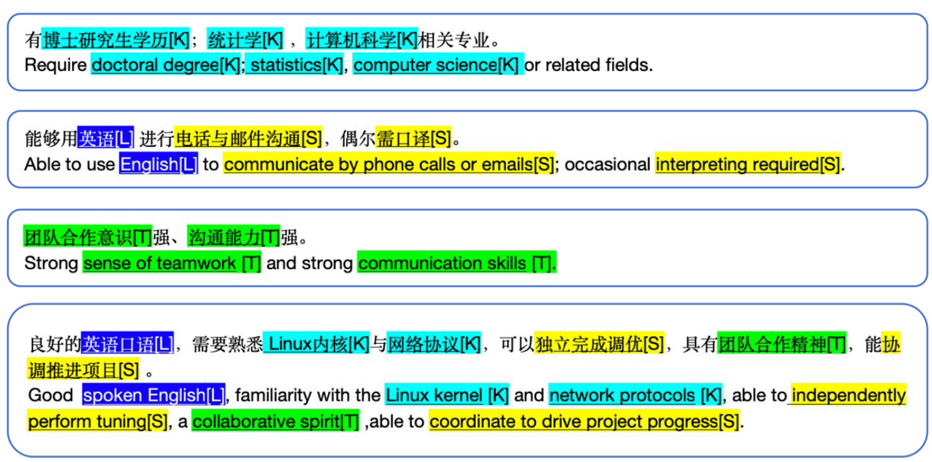

Figure 3. Representative examples of span boundaries and label assignment under the flat, non-overlapping LSKT scheme.

Pre-Release Benchmark Results

Table 2 reports the current evaluation under the standardized annotation and boundary protocol. The two JobBERT-zh systems are supervised encoders trained with protocol-aligned annotations, whereas the language models use the same zero-shot extraction prompt. The 3M JobBERT-zh encoder obtains the highest typed exact score (0.4331), and the 1M encoder obtains the highest typed relaxed score (0.5952). Among the prompted language models, `gpt-5.4` has the highest exact and relaxed scores (0.2132 and 0.4199). These differences should be interpreted in light of supervision: the encoders are trained directly on protocol-aligned spans, while

Table 2. Pre-release results on the standardized Chinese-SkillSpan benchmark. All systems are evaluated on the same 2,601 identifiers using typed exact and typed relaxed micro-F1. Encoder systems use protocol-aligned supervised training; language models use the same zero-shot extraction prompt. Values will be recomputed after the human audit is frozen.

System	Evaluation setting	Typed exact	Typed relaxed
JobBERT-zh, 3M DAPT	Protocol-aligned supervised encoder	**0.4331**	0.5873
JobBERT-zh, 1M DAPT	Protocol-aligned supervised encoder	0.4272	**0.5952**
`gpt-5.4`	Standardized zero-shot prompt	0.2132	0.4199
`deepseek-v4-pro`	Standardized zero-shot prompt	0.1980	0.3931
`kimi-k2.6`	Standardized zero-shot prompt	0.1979	0.4032
`claude-sonnet-4-5`	Standardized zero-shot prompt	0.1972	0.3987
`Qwen2.5-14B-Instruct`	Standardized zero-shot prompt	0.1724	0.3279
`Llama-3-8B-Instruct`	Standardized zero-shot prompt	0.0582	0.1178

the language models receive instructions only. All values remain provisional until the human audit is frozen.

Figure 4 provides the corresponding metric-level comparison under the same V4 matched pre-release protocol.

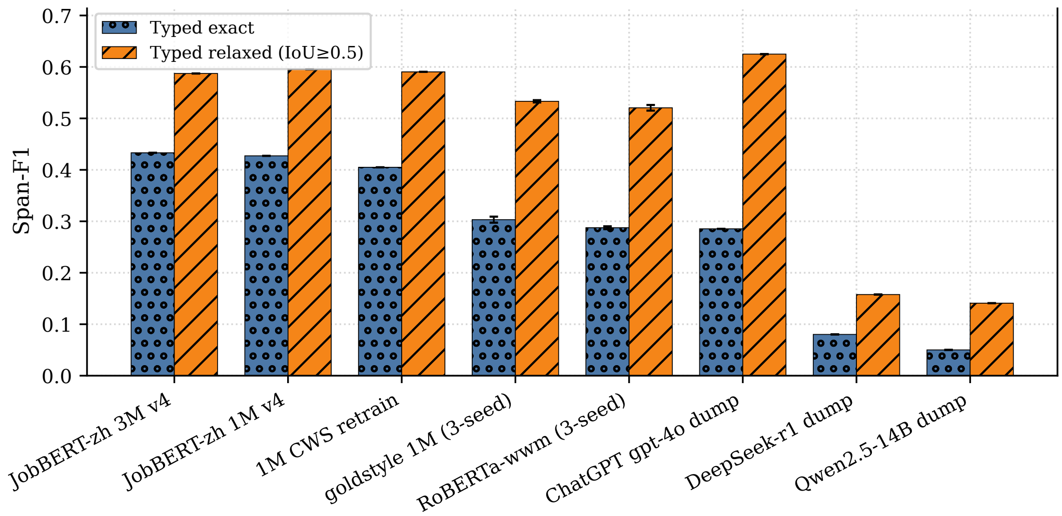

Figure 4. Performance on the V4 matched pre-release protocol, following the layout of SkillSpan Figure 3 (2,601 identifiers; Jieba alignment applied to the reference view and predictions; scorer `cnss-lskt-1.2.0`). Bars report typed exact and typed relaxed micro-F1 at IoU ≥ 0.5. Error bars are sample standard deviations over three seeds where available; V4 JobBERT-zh and frozen language-model dumps are single runs. Within this protocol, JobBERT-zh 3M has the highest typed exact score (0.4331), while frozen `gpt-4o` has the highest typed relaxed score (0.6249). These values are not ranked against the Gold v2 `gpt-4o` score of 0.6365. No SpanBERT-from-scratch or STL/MTL encoder grid was run.

Interpretation and Protocol Sensitivity

The main comparison evaluates two different use cases rather than claiming a protocol-independent ranking of model families. The supervised encoders directly learn the released span convention, while the language models perform zero-shot extraction from written instructions. Their stronger exact-span performance is therefore consistent with protocol-aligned supervision and does not establish universal superiority over language models. The gaps between exact and relaxed scores also show that a substantial portion of the remaining error concerns boundary placement or LSKT type assignment rather than complete semantic failure.

Historical results under the earlier annotation convention, frozen outputs produced with earlier prompts, source-domain analyses, repeated-seed encoder results, and cross-protocol sensitivity checks are reported in the appendix. They are retained for provenance and to quantify the effect of annotation conventions, but they are not combined with Table 2 as a single leaderboard.

RELATED WORK

Span-level skill extraction and typed resources. Research on job-skill mining has gradually evolved from lexicon matching and document-level classification to *span-level* extraction with explicit boundary decisions and ontology linkage. In English, *SkillSpan* established expert-annotated hard- and soft-skill spans and demonstrated the value of boundary-sensitive benchmarking for recruitment text mining (Zhang et al., 2022b). In Danish, *Kompetencer* further highlighted the importance of ESCO-based supervision and the distinction between boundary detection and concept assignment in job-related competency extraction (Zhang et al., 2022a). Recent surveys have summarized the remaining challenges in labor-market NLP, including ontology alignment, long-tail terminology, and domain shift (Senger et al., 2024). Despite this progress, there is still no publicly available Chinese benchmark that provides comparable *span-level*, LSKT-typed competency annotations for recruitment texts.

Table 3 contrasts the annotation level, supervision, scale, label inventory, and release status of the closest span-level resources.

Table 3. Span-level skill-extraction resources. Size denotes annotated data rather than domain-adaptive pretraining data. ✓ indicates public release with the paper. This work's current V4 test reference is a mixed-provenance matched pre-release artifact on the same 2,601 identifiers as Gold v2; the two references are not interchangeable.

Work	Ann.	Approach	Size	Types	Models	Data	Guide.
Sayfullina et al. (2018)	Span	Crowdsourcing	4,863 sent.	Soft	CNN/LSTM	✓	×
Gnehm et al. (2021)	Span	Manual	∼3,000 sent.	Hard	BERT	✓	×
Zhang et al. (2022)	Span	Domain experts	391 JPs / 14.5K sent.	Both (nested)	JobBERT	✓	✓
This work	Span	LLM silver + Doccano	17,460 train sent.; 2,601 test IDs	LSKT (flat)	JobBERT-zh, LLMs	×[†]	✓

[†]Private backup at submission; guidelines in the paper / SI (Handbook B).

Chinese NER benchmarks and domain-specific challenges. Chinese NER research has produced influential benchmarks for social media, newswire, and fine-grained entities, while multilingual evaluations such as MultiCoNER V2 further highlight robustness challenges under noisy and domain-shifted conditions (Peng and Dredze, 2015; Xu et al., 2020; Bhardwaj et al., 2023). Surveys also identify persistent difficulties in Chinese NER, including segmentation ambiguity, character-level granularity, and domain sensitivity (Li et al., 2022). Recent work extends Chinese NER to multimodal and specialized settings (Ji et al., 2024); a generative framework for Chinese legal NER likewise emphasizes the joint difficulty of entity boundaries and fine-grained types in domain-specific text (Mao et al., 2024). However, these datasets primarily target generic or legal entities rather than *competency spans* with an LSKT type system. They therefore cannot directly benchmark ESCO-derived LSKT span extraction in Chinese recruitment texts.

Modeling paradigms for NER and information extraction. From a modeling perspective, recent work has explored both span-based and instruction-based paradigms for entity extraction. Span-oriented methods such as GlobalPointer and SpanMarker have shown strong boundary sensitivity and labeling efficiency compared with BIO-style token tagging, including in Chinese settings (Su, 2021; Aarsen, 2023). At the same time, multilingual encoders such as XLM-R provide useful transferability but remain vulnerable to domain mismatch and ontology drift in specialized recruitment data (Conneau et al., 2020). More recently, open-type and instruction-based extraction frameworks, including GLiNER, InstructUIE, and UniversalNER, have demonstrated that LLM-style or distilled instruction-following models can support flexible zero-shot and few-shot extraction across domains and languages (Zaratiana et al., 2024; Wang et al., 2023; Zhou et al., 2024). These advances suggest that Chinese job-skill extraction should be treated not merely as token classification, but as a boundary-sensitive information extraction task requiring both semantic flexibility and stable type assignment.

Ontologies and evaluation protocols. ESCO provides a hierarchical and cross-lingual vocabulary of skills, competences, qualifications, and occupations, and has become an important reference for labor-market intelligence and cross-lingual skill analysis (Commission, 2024a). Prior span-level job-skill studies emphasize separate evaluation of boundary identification and span typing (Zhang et al., 2022b,a). Motivated by this distinction, we adopt a **flat, non-nested LSKT** type scheme with minimal-complete span boundaries. The present benchmark evaluates ESCO-derived LSKT span extraction.

LLM-assisted annotation and scalable dataset construction. Recent human-in-the-loop annotation pipelines have shown that LLM-generated draft labels, when combined with expert adjudication, can improve annotation efficiency while retaining an auditable review trail. In such settings, annotation quality can be monitored using span-level agreement and token-level consistency statistics. Building on this general paradigm, our workflow uses LLM-generated draft spans followed by rule-based consistency checks and human validation. Model outputs are treated as annotation drafts rather than as independent votes or Gold labels. This design is particularly suitable for Chinese recruitment texts, where span boundaries and competency types often depend on wider contextual interpretation.

Positioning. Against this background, we position *Chinese-SkillSpan* as a Chinese *span-level* competency resource for recruitment texts that combines a **flat ESCO-derived LSKT label**

scheme with **minimal-complete span boundaries**. This scope complements existing English and Danish resources while focusing the evaluation on boundary detection and LSKT type assignment.

LIMITATIONS AND RESPONSIBLE USE

Chinese-SkillSpan is limited to the recruitment sources and domains represented in the collected corpus and should not be interpreted as a complete description of the Chinese labor market. Platform-specific writing conventions and repeated advertisement templates may affect both label distributions and model estimates. The current benchmark evaluates ESCO-derived LSKT span extraction at the category level. The preliminary agreement sample is also too small to establish reliability for rare classes.

Language-model draft annotations can introduce anchoring and model-family biases even when humans review the output. A 980-record subset of the standardized benchmark is still undergoing final manual verification, while the remaining 1,621 records retain protocol-aligned draft annotations with explicit provenance. We therefore describe all current benchmark scores as provisional and reserve *Gold* for the frozen human-adjudicated release. Consolidated agreement and adjudication statistics will be reported after the manual audit is complete. The dataset and derived models are intended for research on information extraction and aggregate labor-market analysis. They should not be used without additional validation and human oversight to rank applicants, make hiring decisions, or infer individual capability. Any public release must also document source permissions, redistribution constraints, and the removal of personal or contact information.

CONCLUSION

We introduced **Chinese-SkillSpan**, a 22,840-sentence Chinese span-level competency resource for job advertisements with an ESCO-derived LSKT type system and Chinese-specific boundary guidelines. The release centers on an identifier-aligned standardized benchmark, documented annotation rules, a reproducible scorer, and analyses across recruitment sources, span lengths, model families, and random seeds.

The experiments show that exact-span scores depend strongly on boundary conventions and source domain. In particular, model ordering can change when encoder supervision, decoding boundaries, and the evaluation reference follow the same operational convention. This finding should be interpreted as protocol sensitivity rather than evidence that one model family is universally superior.

Before submission, the ongoing 980-record manual audit must be adjudicated and frozen, after which all main benchmark tables will be regenerated from one manifest. Historical results under earlier annotation conventions will remain in the appendix as provenance rather than as a competing benchmark. Remaining release tasks include recording model access metadata and checksums and publishing the versioned data, scorer, and code with clear governance documentation. Chinese-SkillSpan provides a transparent foundation for studying Chinese job-skill extraction and for evaluating how operational annotation decisions affect exact-span benchmarking.

SUPPLEMENTARY ANNOTATION MATERIALS

This appendix presents the active-draft LSKT-v4 annotation manual. The V4 reference is undergoing final human verification and adjudication; therefore, the tables define the operational decisions but do not imply that the V4 audit or IAA has already been completed. The design follows useful elements of the SkillSpan manual—iterative calibration, explicit boundary exceptions, overlapping independent annotation, adjudication, and rechecking after guideline revisions (Zhang et al., 2022b)—while adapting them to Chinese segmentation and the flat LSKT inventory. Unlike SkillSpan, Chinese-SkillSpan does not permit nested spans. Tables 4 and 5 present the general rules; Tables 6, 7, 8, and 9 present type decisions, conflict resolution, annotation and derived-view formats, and the V4 audit plan. The complete transition from the Gold-era rules to SOP-v4, including deprecated rules, is recorded in `handbooks/LSKT_V4_RULE_CHANGELOG.md`. Vector embedding keeps the Chinese glyphs sharp and independent of the manuscript compiler.

General Annotation Guidelines

Chinese SkillSpan SOP-v4 Guidelines (A1, Part I: G1-G7)

Active draft; human verification and adjudication are in progress.

Rule ID	Operational decision	Chinese example
G1	Use exactly four labels: L (language competence), K (knowledge), S (specific or operational skill), and T (transversal competence). Keep them distinct during annotation.	标注阶段只使用 L/K/S/T；任何二分类映射仅用于后续评测。
G2	Use flat, non-overlapping spans. A character position belongs to at most one labeled span.	同一表层片段最多一个标签；不允许重叠、嵌套或跨跨度重复标注。
G3	Select an exact contiguous substring of the original sentence. Never rewrite a mention into a canonical form; store normalization separately.	原句：熟悉 Jekins 标注：@@Jekins##[S]（保留原文拼写，不改为 Jenkins）。
G4	Select the shortest complete mention and never cut through a word or identifier. Two to eight segmented tokens is a preference, not a hard cap.	保留：@@支持服务##[S] 禁止：@@支持服##[S]。完整长跨度不得为满足长度偏好而截断。
G5	For 熟悉/掌握/精通/了解/具备/具有, exclude the competence trigger and label its object. Retain a predicate only when it is part of an executable skill.	熟悉 Python -> @@Python##[S]；掌握 SQL 原理 -> @@SQL原理##[K]。
G6	Split coordinated expressions only when each conjunct independently names a competency. Keep one span when conjuncts share an indispensable head, object, or complement.	Python 与 Java -> @@Python##[S]；@@Java##[S] 维护和支持服务 -> @@维护##[S]；@@支持服务##[S]。
G7	Retain any head, object, or complement required to make the competency self-contained.	标注：@@协议栈配置##[S]；@@端到端部署##[S]；@@A/B测试设计##[S]。

Table 4. Draft SOP-v4 general annotation guidelines, A1 Part I (G1–G7). The panel defines exact source substrings, shortest-complete boundaries, trigger exclusion, conditional coordination splitting, and necessary complements.

Chinese SkillSpan SOP-v4 Guidelines (A1, Part II: G8-G14)

Active draft; Jieba is a validator/derived view and does not overwrite human-frozen offsets.

Rule ID	Operational decision	Chinese example
G8	Return an empty annotation for application logistics, medical examinations, public notices, benefits, locations, schedules, company introductions, slogans, and vague non-competency duties.	五险一金，带薪年假 -> 空标注 []；体检时间另行通知 -> 空标注 []。
G9	Exclude generic scene, degree, frequency, time, attitude, and pronoun modifiers unless removing them changes the competency identity or L/K/S/T type.	通常排除：良好、三年以上、相关、该、其、本；若修饰语改变类型则提交仲裁。
G10	Natural-language names, proficiency levels, and language examinations are L. Technical standards and professional certifications are K.	@@英语六级##[L]；@@CET-6##[L]；@@ISO 27001##[K]；@@OCJP-Java认证##[K]。
G11	Programming languages, frameworks, platforms, software, and tools required for job use are S. Explicit theory or principles are K.	@@Python##[S]；@@Excel##[S]；@@数据库原理##[K]。
G12	Use predicate and referent context for SQL and other ambiguous technologies. Do not label an entire responsibility clause as one S span.	使用SQL进行报表制作 -> @@SQL##[S]；@@报表制作##[S] 掌握SQL原理 -> @@SQL原理##[K]。
G13	Do not apply a global label priority. Record the sentence, candidate spans, candidate labels, applicable rules, and rationale for adjudication.	未决 K/S、L/S、L/K 或 T/S 案例进入仲裁，不使用 L>S>K>T。
G14	Use zero-based, end-exclusive character offsets in the original sentence. Human-frozen offsets are authoritative; BIO and CWS-aligned views are derived artifacts.	校验：sentence[start:end] 必须与 span 完全一致；Jieba 不得自动覆盖人工 Gold。

Table 5. Draft SOP-v4 general annotation guidelines, A1 Part II (G8–G14). CET-6 is L; technical certifications are K; programming languages and tools are S in executable-use contexts; SQL is typed by predicate context; Jieba does not overwrite human-frozen offsets.

ESCO Alignment Criteria

ESCO-derived LSKT Type Decisions (SOP-v4 Draft)

CET-6 is L; programming languages and tools are S in executable-use contexts.

Label	Operational definition	Chinese example
L - Language	Natural-language proficiency, level, or language examination/certificate.	英语；商务英语；@@英语六级##[L]；@@CET-6##[L]；@@日语N2##[L]
K - Knowledge	Education, major, domain knowledge, theory, technical standard, or professional certification.	计算机相关专业；数据库原理；@@ISO 27001##[K]；@@OCJP-Java认证##[K]
S - Specific skill	Executable occupation-specific action, method, programming language, framework, platform, software, or tool.	@@Python##[S]；@@Excel##[S]；数据清洗；模型训练；接口对接；报表制作
T - Transversal	Transferable cross-role competence or disposition.	沟通能力；团队协作；责任心；抗压能力
Context test	Language qualification -> L; technical certification/theory -> K; technology used to execute work -> S; general cross-role trait -> T.	英语六级[L]；SQL原理[K]；使用SQL中的 SQL[S]；沟通能力[T]

Table 6. SOP-v4 LSKT type decisions derived from high-level ESCO distinctions. The four labels remain distinct during annotation; any collapsed projection is evaluation-only.

Conflict Resolution

SOP-v4 Conflict Scenarios & Resolution Rules

Context replaces the legacy global label priority.

Conflict type	Decision rule	Chinese example
K <-> S	Explicit theory, principles, or standards -> K; executable use, action, tool, or output -> S.	掌握@@SQL原理##[K]；使用@@SQL##[S]进行@@报表制作##[S]
L <-> K	Natural-language proficiency/examination -> L; technical standard or professional certification -> K.	@@英语六级##[L]；@@CET-6##[L]；@@OCJP-Java认证##[K]
L <-> S	A language ability is L; a concrete language-mediated work action is S. Distinct surfaces may both be labeled.	能用@@英语##[L]与海外客户@@邮件沟通##[S]
T <-> S	A general cross-role trait -> T; a concrete action or deliverable -> S.	具备@@沟通能力##[T]；主动@@协调项目进度##[S]
Coordination	Split independent competency mentions; keep one span when conjuncts share an indispensable head/object/complement.	@@Python##[S]与@@Java##[S]；@@维护##[S]和@@支持服务##[S]
Unresolved	Do not force a global L>S>K>T priority. Send unresolved type or boundary cases to adjudication with a written rationale.	记录：原句、候选跨度、候选标签、适用规则、最终决定与仲裁人。

Table 7. Contextual conflict-resolution rules for K/S, L/K, L/S, and T/S decisions. Unresolved cases are adjudicated with a written rationale rather than forced through a global label priority.

Annotation Format

SOP-v4 Annotation and Derived-View Format

Human-frozen character offsets are the source of truth; BIO and CWS files are reproducible derivatives.

Rule ID	Requirement	Chinese example
4.1	Authoritative record: sentence ID, original sentence, zero-based start, end-exclusive end, exact span text, and one L/K/S/T label.	sentence[start:end] == span 必须为真。
4.2	The extracted text is an exact source substring. Normalized names or ontology links, when available, are stored in a separate field.	原文 Jekins -> @@Jekins##[S]；不得在 span 中改写为 Jenkins。
4.3	One span carries exactly one label; spans are flat and non-overlapping.	@@数据可视化##[S]；@@数据库原理##[K]；@@沟通能力##[T]；@@CET-6##[L]
4.4	BIO sequences are generated only after character spans are frozen and must round-trip to the same offsets.	BIO -> span -> BIO 往返后，边界和 L/K/S/T 标签必须不变。
4.5	Jieba is a boundary validator and named derived-view transform, not the human annotation authority.	检测到半词边界 -> 返回人工复核；不得自动覆盖人工 Gold。
4.6	A CWS-derived evaluation view freezes Jieba version, user dictionary, rewrite function, cap, and checksum, and names the transformed reference and prediction files.	同一派生评测中，reference 与 prediction 使用同名、同版本转换。
4.7	Model JSON output is validated against the same manual before offsets are derived; missing, extra, duplicated, or non-verbatim records fail validation.	输出 text 必须在对应 sentence 中原样出现；所有 ID 顺序与输入一致。

Table 8. SOP-v4 annotation and derived-view format. Human-frozen character-offset JSONL is authoritative; normalization, BIO sequences, and any CWS-aligned evaluation file are separately named derived artifacts.

Quality Control and Output

SOP-v4 Quality Control and IAA Plan

The V4 IAA study is planned but not yet complete; historical n=100 statistics are not reused as V4 evidence.

Stage	Requirement	Status / evidence
Current status	Keep the V4 reference provisional until final human verification, adjudication, manifests, and checksums are frozen.	V4 人工核对进行中；当前 V4 数值必须标为 provisional。
Blind sample	Select 300 records from the 980-record verification pool with documented source-domain, sentence-length, and difficulty strata.	两名标注员只看原始文本，不看模型草稿、旧 Gold 或彼此结果。
Pre-adjudication IAA	Compute typed exact span P/R/F1, typed relaxed F1 at IoU >= 0.5, character-level Cohen's kappa over O/L/K/S/T, and per-label scores before adjudication.	先冻结 A/B 独立文件和指标，再开始仲裁；不预设 kappa 达标阈值。
Adjudication	A third annotator resolves disagreements and records the sentence, competing spans/labels, applicable rules, final decision, and rationale.	若规则变化，回查所有受影响记录并更新 change log。
Historical evidence	The published n=100 agreement figures describe Gold-era boundaries only and must not be reported as V4 IAA.	历史：strict F1=0.5320；relaxed F1=0.6240；kappa=0.5540（非 V4）。
Traceability	Freeze guideline version, prompt hash, annotator IDs, timestamps, sample manifest, Jieba/configuration manifest for derived views, scorer version, and checksums.	缺失的历史元数据标记为 missing，不推测补写。
Output gate	Freeze human character-offset JSONL first; derive BIO and named CWS views afterward and verify round-trip consistency.	最终输出：[id，sentence，start，end，span，label，provenance]。

Table 9. SOP-v4 quality-control and IAA plan. The proposed 300-record dual-blind study is pending; the historical 100-sample figures describe the Gold-era convention and are not reused as V4 agreement evidence.

The historical 100-sample overlap audit reported strict span F1 of 0.5320, relaxed span F1 of 0.6240, and token-level Cohen's κ of 0.5540 under the Gold-era convention. The V4 study is separate: two annotators will label the same stratified 300 raw-text records without access to model drafts or prior spans; agreement will be logged before adjudication. Until those artifacts are frozen, no V4 IAA value is claimed.

WORKED ANNOTATION EXAMPLES

Table 10 provides author-prepared Chinese contrast examples for the five frozen decisions and the remaining core boundary rules. Corpus-derived examples must additionally carry a released source identifier; constructed examples are identified as author-prepared rather than presented as corpus observations.

SOP-v4 Quick Worked Examples

Examples retain Chinese surface text; any corpus-derived release example must also carry a source ID in the provenance file.

ID	Decision illustrated	Author-prepared Chinese example and tags
E1	Language qualification is L; technical certification is K.	原句：大学英语六级，通过ISO 27001认证 标注：@@大学英语六级##[L]；@@ISO 27001##[K]
E2	Competence triggers are excluded; tools and actions are S.	原句：熟练使用Excel进行数据汇总 标注：@@Excel##[S]；@@数据汇总##[S]
E3	Theory is K; executable programming/tool requirements are S.	原句：掌握数据库原理，熟悉Python与PyTorch 标注：@@数据库原理##[K]；@@Python##[S]；@@PyTorch##[S]
E4	SQL is typed by explicit predicate context.	原句：使用SQL进行报表制作，掌握SQL原理 标注：@@SQL##[S]；@@报表制作##[S]；@@SQL原理##[K]
E5	Independent coordination is split; necessary complements are retained.	原句：协议栈配置、端到端部署及A/B测试设计 标注：@@协议栈配置##[S]；@@端到端部署##[S]；@@A/B测试设计##[S]
E6	Language ability and language-mediated work are distinct spans.	原句：能用英语与海外客户邮件沟通 标注：@@英语##[L]；@@邮件沟通##[S]
E7	General transferable competence is T; concrete output is S.	原句：具备沟通能力，能够编写测试用例 标注：@@沟通能力##[T]；@@编写测试用例##[S]
E8	Administrative and benefit-only text receives no competency span.	原句：体检时间另行通知，入职后享受五险一金 标注：[]

Table 10. Worked SOP-v4 examples covering language qualifications, technical certifications, programming tools, SQL context, coordination, L/S and T/S contrasts, and empty administrative or benefit text.

EXPERIMENTAL-SCOPE INVENTORY

Table 11 makes the experimental scope explicit without placing incomparable F1 values side by side. SkillSpan provides the stronger repeated-run encoder design, whereas Chinese-SkillSpan covers more annotation types, protocol variants, and LLM settings.

Table 11. Scope comparison with SkillSpan (Zhang et al., 2022b). Counts describe resources and evaluation dimensions, not a common performance leaderboard.

Dimension	SkillSpan	Chinese-SkillSpan
Annotated resource	391 English job postings; 14.5K sentences	22,840 Chinese sentences; 17,460/2,143/3,237 train/dev/test
Label inventory	Two nested labels: skill and knowledge	Four flat, non-overlapping labels: L/K/S/T
Unlabeled adaptation corpus	3.2M job-ad sentences	1M and 3M Chinese job-ad DAPT conditions
Encoder scope	BERT, SpanBERT, JobBERT, JobSpanBERT; STL/MTL	RoBERTa-wwm, JobBERT-zh 1M/3M, domain-mix, listed-mix; CRF
Repeated runs	Five seeds with ASO significance tests	Three seeds with mean and sample standard deviation
Additional systems	Longformer negative result	Frozen LLM suite and six same-prompt LLM diagnostics
Error analysis	Source, P/R, prediction length, span length	Source, P/R, span length, protocol and boundary-snap diagnostics
Evaluation references	One expert-annotated test reference	One standardized pre-release benchmark plus archived historical references

REPRODUCIBILITY AND PROTOCOL INVENTORY

Table 12 records the internal artifact associated with each protocol layer. The main text uses professional descriptive terms, whereas this appendix preserves the exact P0/P1/P2 and V4 identifiers required to reproduce the analysis. The standardized P2 reference is the current pre-release benchmark; P0 and P1 are retained only for historical reproduction and protocol-sensitivity analysis.

HISTORICAL ANNOTATION AND EVALUATION RESULTS

The tables in this section document results produced under earlier annotation conventions. They are preserved to make the dataset's provenance auditable and to support protocol-sensitivity analysis. They are not alternative headline results and must not be combined with the standardized benchmark table in the main text.

PRECISION AND RECALL ON GOLD V2

Table 19 shows that the encoder baselines have substantially lower recall than precision. The Qwen dump is also high-precision/low-recall, whereas `gpt-4o` is the only system with balanced high precision and recall.

DOMAIN STABILITY ACROSS SEEDS

Table 20 confirms that random-seed variation does not explain the encoder failure on public-sector recruitment: the three-seed means remain between 0.0126 and 0.0234, compared with 0.7032 for `gpt-4o`.

Figure 5 visualizes the same three-seed means. The uniformly light public-sector encoder cells make the source-specific collapse visible, while the frozen `gpt-4o` row remains strong across all three subsets.

Table 12. Mapping between manuscript terminology and internal versioned artifacts. These exact identifiers are retained for reproducibility but are not used as narrative benchmark names in the main text.

Component	Artifact	Coverage	Permitted role
Historical P0 reference	`admin_Baseline_test.jsonl`	2,676 rows	Original paper reproduction only
Legacy P1 reference (Gold v2)	`gold_canonical_v2.jsonl`	2,601 unique-first IDs	Appendix provenance and sensitivity analyses only
Standardized P2 pre-release reference	`test_lskt_v4_cws_simhuman980_hybrid.jsonl`	Same 2,601 IDs; 980-record audit in progress	Current main benchmark; provisional until human adjudication
Protocol-aligned training labels (SOP-v4)	`train_lskt_v4_silver`	Training split only	Encoder supervision; never used as test Gold
Official scorer	`cnss-lskt-1.2.0`	ID-strict alignment	Typed exact micro-F1; relaxed IoU ≥ 0.5 as a secondary metric
Model inventory	`tables/model_ids.csv`	Frozen model names and coverage status	Links each result row to its evaluated model endpoint or checkpoint

Table 14. Historical inter-annotator agreement on 100 samples under the earlier annotation convention. These values do not measure agreement under the standardized benchmark protocol.

Agreement metric	Score
Strict exact-span F1	0.5320
Relaxed span F1	0.6240
Token-level Cohen's κ	0.5540

Table 15. Legacy Gold v2 unique-first results on 2,601 identifiers using `cnss-lskt-1.2.0`. Columns report typed exact micro-F1, collapsed exact micro-F1, and typed relaxed micro-F1 at IoU ≥ 0.5.

System	Typed exact	Collapsed exact	Typed relaxed	Coverage/status
`gpt-4o`	**0.6365**	**0.6403**	**0.7221**	Complete
`claude-3-5-haiku-20241022` (filled)	0.2583	0.2970	0.3861	98 Sonnet 4.6 fills
`kimi-k2-0711-preview`†	0.1651	0.3349	0.2130	293 IDs missing
`deepseek-r1`	0.1327	0.3569	0.1798	Complete
`Qwen2.5-14B-Instruct`	0.0791	0.1075	0.1272	Complete; paper-score gap
JobBERT-skill	0.0000	0.0045	0.0000	English head
JobBERT-knowledge	0.0000	0.0037	0.0000	English head

†The incomplete Kimi row is diagnostic and is not ranked against complete systems.

Table 17. Top-5 Skill (S+T) and Knowledge (L+K) surface strings on **Gold v2** (Doccano provenance; the same 2,601 identifiers used by V4). The noisier V4 hybrid frequencies are intentionally excluded from this descriptive table.

	AI	Cloud	Public
SKILL	开发; 设计; 分析; 沟通; 优化	维护; 沟通能力; 沟通; 优化; 运维	面试; 体检; 考试; 加试; 科研能力
KNOW.	计算机; 云计算; 大数据; 机器学习; 算法	计算机; 网络; 服务器; 计算机相关专业; ERP	综合成绩; 体检标准; 笔试; 公共基础知识; 成绩

Table 18. Historical paper-reported S-F1 results under the raw 2,676-row protocol (original PDF Table 3).

System	Paper S-F1
ChatGPT (`gpt-4o`)	0.6700
Claude (`claude-3-5-haiku-20241022`)	0.6300
Kimi (`kimi-k2-0711-preview`)	0.5700
DeepSeek (`deepseek-r1`)	0.5130
Qwen (`Qwen2.5-14B-Instruct`)	0.2130
JobBERT-skill	0.0045
JobBERT-knowledge	0.0038

SPAN-LENGTH ANALYSIS

The mean Gold v2 span length is 4.90 tokens. Table 21 shows that `gpt-4o` is weak on length-one spans but strong on lengths two and four. Both seed-42 encoders predict spans that are too long on average and perform poorly at lengths one and ten or more.

Figure 8 separates two reference protocols instead of treating their scores as one ranking. On Gold v2, the frozen `gpt-4o` dump remains above the Gold-style encoders for most span lengths. This pattern is consistent with closer alignment between the draft model and the Gold v2 boundary convention, but it does not establish that human decisions were copied from `gpt-4o`. In the pre-audit matched SOP–CWS snapshot, JobBERT-zh trained with SOP labels is higher than the same frozen `gpt-4o` dump in every populated length bin (2–8 tokens). Together with Table 27, this reversal is consistent with annotation and boundary compatibility contributing to exact-span performance; it does not isolate that contribution from simultaneous changes in supervision and decoding. The panel will be regenerated after the 980-record audit is frozen.

Figures 6 and 7 provide complementary Gold v2 diagnostics for predicted span length and typed exact performance by reference span length.

Table 19. Typed exact precision, recall, and F1 on Gold v2 using `cnss-lskt-1.2.0`. Encoder entries are three-seed means ± sample standard deviations. The Kimi row empty-fills 293 missing IDs and is not a complete-coverage result.

System	Precision	Recall	F1
ChatGPT (`gpt-4o`)	0.6264	0.6469	0.6365
Claude filled (Haiku + 98 Sonnet 4.6)	0.2300	0.2947	0.2583
JobBERT-zh 1M Gold-style v3	0.1864 ± 0.0137	0.0984 ± 0.0037	0.1288 ± 0.0062
Domain-mix 1M	0.1841 ± 0.0029	0.0969 ± 0.0033	0.1269 ± 0.0031
JobBERT-zh 3M checkpoint 65000	0.1785 ± 0.0055	0.0972 ± 0.0030	0.1258 ± 0.0033
RoBERTa-wwm v3	0.1695 ± 0.0126	0.0928 ± 0.0024	0.1199 ± 0.0050
DeepSeek (`deepseek-r1`)	0.1384	0.1274	0.1327
Kimi empty-fill (293 missing)	0.1677	0.1393	0.1522
Qwen (`Qwen2.5-14B-Instruct`)	0.2178	0.0483	0.0791

Table 20. Gold v2 typed exact F1 by source domain. Encoder values are three-seed means ± sample standard deviations; the domain sizes are 1,407, 457, and 737 IDs.

System	AI recruitment	Alibaba Cloud	Public-sector
ChatGPT (`gpt-4o`)	0.6489	0.5650	0.7032
JobBERT-zh 1M	0.1365 ± 0.0081	0.1354 ± 0.0039	0.0213 ± 0.0034
Domain-mix 1M	0.1334 ± 0.0052	0.1352 ± 0.0021	0.0234 ± 0.0071
JobBERT-zh 3M	0.1344 ± 0.0021	0.1315 ± 0.0078	0.0147 ± 0.0007
RoBERTa-wwm v3	0.1301 ± 0.0072	0.1186 ± 0.0025	0.0126 ± 0.0014

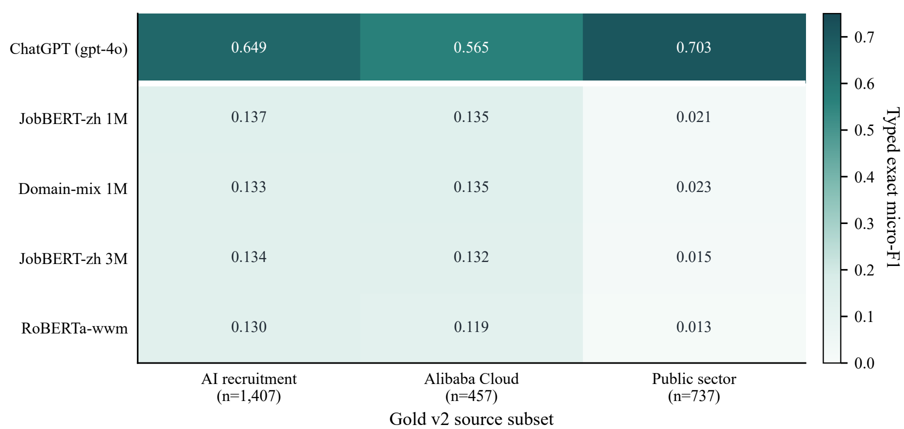

Figure 5. Source-domain robustness on Gold v2. Encoder cells report three-seed mean typed exact micro-F1, while `gpt-4o` is a single frozen-dump result. The horizontal separator distinguishes the LLM reference from Chinese encoder baselines.

Table 21. Gold v2 typed exact F1 by gold token length, where length is $end - start$. The last column is the mean predicted span length.

System	1	2	3	4	5	6	7	8	9	10+	Mean pred.
ChatGPT (`gpt-4o`)	0.0357	0.7265	0.5643	0.7292	0.5095	0.6702	0.5066	0.5962	0.5183	0.5054	4.19
JobBERT-zh 1M (seed 42)	0.0000	0.1229	0.1347	0.1923	0.0758	0.1570	0.0669	0.0858	0.0743	0.0357	5.94
RoBERTa-wwm v3 (seed 42)	0.0000	0.1105	0.1146	0.1882	0.0924	0.1357	0.0629	0.0851	0.0890	0.0372	5.98

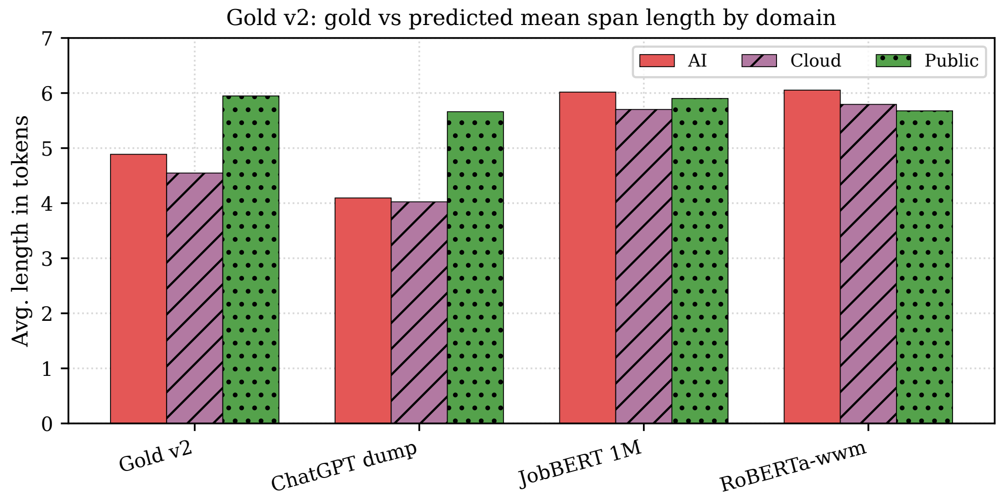

Figure 6. Mean span length on Gold v2, following the layout of SkillSpan Figure 5. The encoders predict longer spans (approximately 5.9–6.1 tokens) than the Gold v2 domain means (approximately 4.5–5.9 tokens), whereas the frozen `gpt-4o` dump is closer to the reference length.

DESCRIPTIVE SEED-WIN MATRIX

Table 22 is a descriptive robustness check rather than a significance test. With only three seeds per encoder, the three domain-adapted variants are not reliably separable; each nevertheless wins more often than RoBERTa-wwm in the nine cross-seed comparisons.

Table 22. Pairwise seed win rates, computed as $P(F1_{\text{row}} > F1_{\text{column}})$ over 3×3 seed pairs. This under-powered descriptive matrix is not the five-seed ASO test used by SkillSpan and carries no significance claim.

Row \ column	JB 1M	Domain	JB 3M	RoBERTa
JobBERT-zh 1M	—	0.5556	0.5556	0.8889
Domain-mix	0.4444	—	0.5556	0.8889
JobBERT-zh 3M	0.4444	0.4444	—	0.7778
RoBERTa-wwm	0.1111	0.1111	0.2222	—

Figure 9 visualizes the same 3×3 pairwise seed comparisons. It is deliberately presented as a descriptive win probability, not as an ASO or hypothesis-test result.

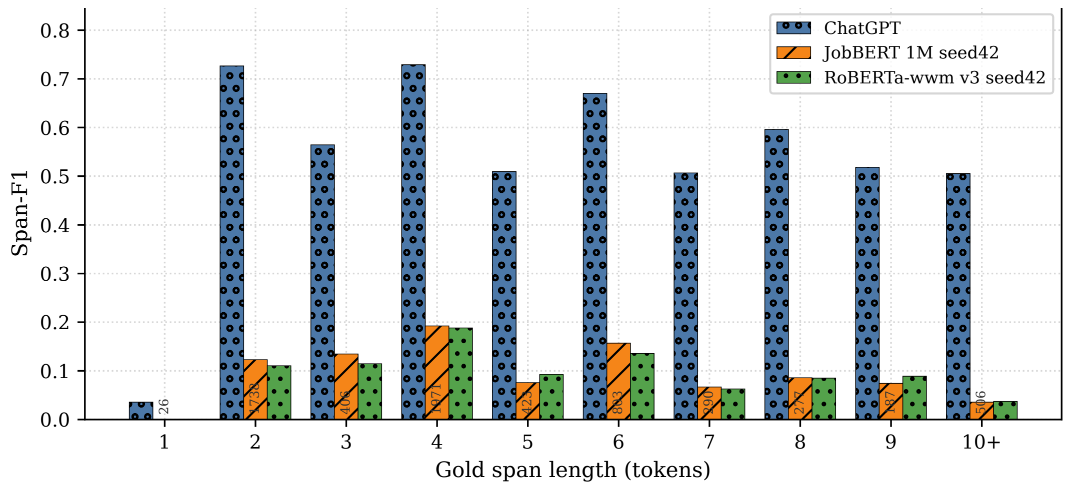

Figure 7. Gold v2 typed exact micro-F1 by reference span length, following the layout of SkillSpan Figure 6. Vertical annotations give the number of Gold v2 reference spans in each bucket. The frozen `gpt-4o` dump is weak on length-one spans but strong at lengths two and four; the two seed-42 encoders remain below 0.20 in every bucket.

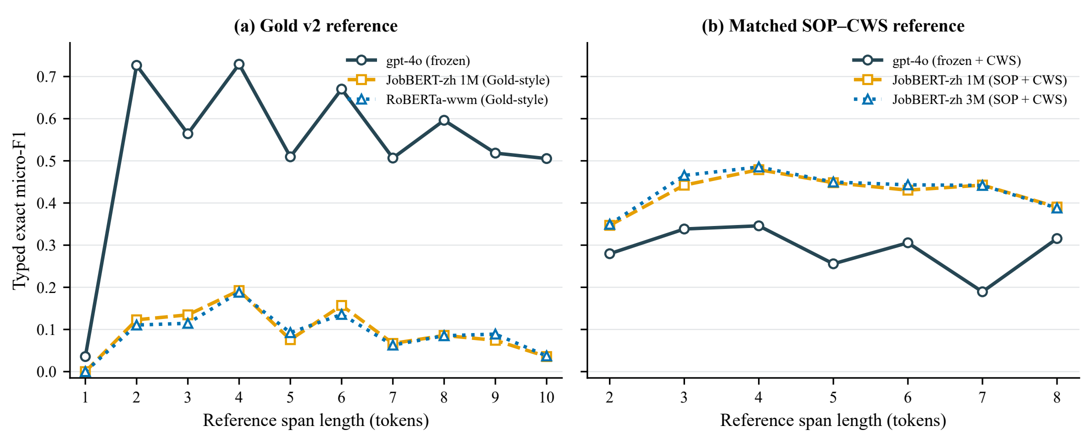

Figure 8. Typed exact micro-F1 by reference span length under two non-interchangeable protocols. (a) Gold v2 uses the original reference and Gold-style encoder training. (b) The pre-audit matched SOP–CWS snapshot uses the same 2,601 IDs and bilateral Jieba boundary snapping; its operational limits yield populated bins only from 2 to 8 tokens. The frozen `gpt-4o` dump is the same source prediction in both panels, with boundary snapping applied only in panel (b). Scores across panels should not be compared as a single leaderboard, and panel (b) will be regenerated after the manual audit.

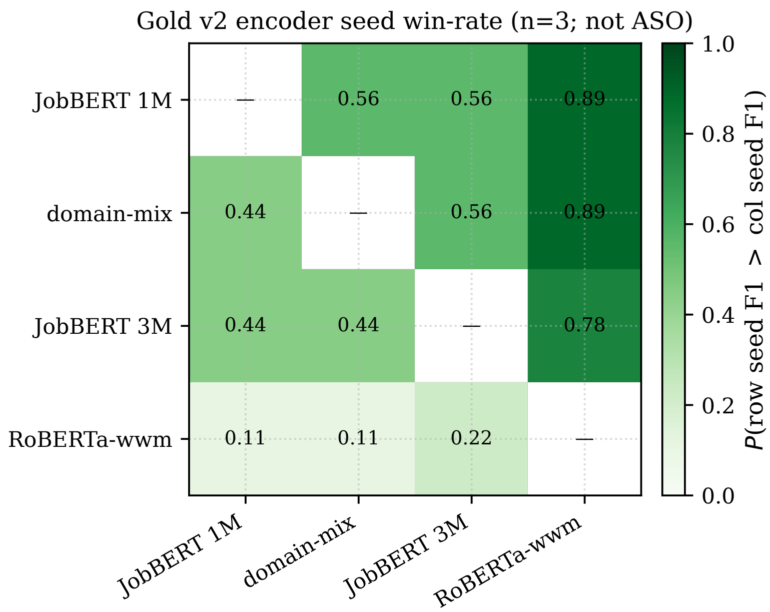

Figure 9. Gold v2 encoder pairwise seed win rate, following the layout of SkillSpan Figure 4. Each cell is $P(F1_{\text{row}} > F1_{\text{column}})$ over 3×3 seed pairs. This descriptive analysis uses $n = 3$ and is not an Almost Stochastic Order test with Bonferroni correction; the 1M, domain-mix, and 3M variants are not reliably separable.

PRECISION AND RECALL ON THE MATCHED PROTOCOL

The values in Table 23 were computed from the current pre-audit matched snapshot. They are retained to document the ongoing analysis and will be recomputed after the 980-record manual audit is adjudicated and frozen.

Table 23. Pre-audit typed exact precision, recall, and F1 on the P2 matched SOP–CWS snapshot. These provisional values must not be combined with the Gold v2 results in Table 19.

System	Precision	Recall	F1
JobBERT-zh 3M v4 + Jieba	0.4730	0.3994	0.4331
JobBERT-zh 1M v4 + Jieba	0.4685	0.3925	0.4272
JobBERT-zh 1M CWS retrain + Jieba	0.4537	0.3655	0.4049
ChatGPT (`gpt-4o`, frozen + Jieba)	0.2371	0.3584	0.2854

PROTOCOL-ALIGNMENT DIAGNOSTICS

Table 24 separates changes in encoder supervision, decoding boundaries, and evaluation references. Exact scores increase when training labels, decoding, and the diagnostic reference implement the same span convention. Because more than one component changes across rows, these values measure protocol compatibility rather than a causal effect of any single operation.

Table 24. Internal SOP-v4 and CWS diagnostics by encoder checkpoint, training annotation source, decoding rule, and test reference. Gold v2, SOP-rule Silver, and SOP–CWS Silver are distinct references and are not interchangeable.

Prediction	Training labels	Decode	Test reference	Typed exact	Typed relaxed
JobBERT 1M	Gold-style v3	Raw	Gold v2	0.1224	—
JobBERT 1M	SOP v4	Raw	Gold v2	0.1079	0.3320
JobBERT 3M	SOP v4	Raw	Gold v2	0.1104	0.3404
JobBERT 1M	SOP v4	Jieba post-hoc	Gold v2	0.1454	0.3411
JobBERT 3M	SOP v4	Jieba post-hoc	Gold v2	0.1479	0.3470
JobBERT 1M	SOP v4	Raw	SOP rule Silver	0.3170	0.5663
JobBERT 3M	SOP v4	Raw	SOP rule Silver	0.3229	0.5624
JobBERT 1M	SOP v4	Jieba post-hoc	SOP rule Silver	0.2609	0.5835
JobBERT 1M	SOP v4	Jieba post-hoc	SOP–CWS Silver	0.4278	0.5960
JobBERT 3M	SOP v4	Jieba post-hoc	SOP–CWS Silver	0.4341	0.5884

Table 25 preserves the frozen outputs produced before the standardized prompt recall. These rows are useful for provenance and boundary-sensitivity analysis, but they are not included in the main benchmark table because the language-model prompts and coverage differ.

CROSS-PROTOCOL DIAGNOSTIC

Table 27 places the confirmed P1 scores beside the provisional P2 snapshot to expose protocol sensitivity. Under P1 Gold v2, the frozen `gpt-4o` dump is higher than the SOP-trained encoders on both exact and relaxed F1. In the current P2 snapshot, the SOP-trained encoders have higher exact F1, while `gpt-4o` retains the highest relaxed F1. The reversal is therefore concentrated in exact boundary-and-type agreement rather than semantic overlap. Because the reference annotations, encoder supervision, and protocol-specific decoding are not all held fixed, the table is diagnostic and does not establish a protocol-independent ranking or a causal effect of any single component.

Table 25. Frozen-output diagnostic on the internal P2 matched SOP–CWS snapshot. The reference comprises 980 records under human verification and 1,621 draft-derived records. Scores are provisional and will be regenerated after adjudication.

System	Typed exact	Typed relaxed	Coverage/status
JobBERT 3M v4 + Jieba	**0.4331**	0.5873	Complete
JobBERT 1M v4 + Jieba	0.4272	0.5952	Complete
JobBERT 1M CWS retrain + Jieba	0.4049	0.5904	Complete
Domain-mix 1M (three-seed mean)	0.3037	0.5278	Complete
JobBERT 1M Gold-style v3 (three-seed mean)	0.3032	0.5332	Complete
Listed-mix 1M	0.2964	0.5267	Complete
JobBERT 3M checkpoint 65000 (three-seed mean)	0.2961	0.5278	Complete
JobBERT demo 80k	0.2931	0.5321	Complete
RoBERTa-wwm v3 (three-seed mean)	0.2875	0.5206	Complete
`gpt-4o` (frozen + Jieba)	0.2854	**0.6249**	Complete; earlier prompt
`claude-3-5-haiku-20241022`	0.1483	0.3349	98 IDs missing
`kimi-k2-0711-preview`	0.0964	0.1997	293 IDs missing
`deepseek-r1` (frozen + Jieba)	0.0802	0.1577	Complete; earlier prompt
`Qwen2.5-14B-Instruct` (frozen + Jieba)	0.0501	0.1409	Complete; earlier prompt
JobBERT-skill English head	0.0096	0.0676	Complete
JobBERT-knowledge English head	0.0088	0.0644	Complete

Table 27. Cross-protocol diagnostic using the confirmed P1 pipeline and the provisional P2 snapshot. P1 is the human-reviewed Gold v2 reference; P2 is the mixed-provenance matched SOP–CWS artifact undergoing a 980-record manual audit. The same frozen `gpt-4o` source dump is evaluated raw under P1 and with Jieba boundary snapping under P2, whereas both JobBERT-zh rows use SOP-v4 training and Jieba decoding in both columns. Exact and relaxed values are typed micro-F1. Between-protocol changes diagnose compatibility and must not be interpreted as a unified leaderboard.

		P1: Gold v2		P2: matched SOP–CWS	
System	**Decode (P1 / P2)**	**Exact ↑**	**Relaxed ↑**	**Exact ↑**	**Relaxed ↑**
`gpt-4o` frozen	Raw / Jieba	**0.6365**	**0.7221**	0.2854	**0.6249**
JobBERT-zh 1M SOP v4	Jieba / Jieba	0.1454	0.3411	0.4272	0.5952
JobBERT-zh 3M SOP v4	Jieba / Jieba	0.1479	0.3470	**0.4331**	0.5873

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